

PRATIK SINGH

AI/ML Engineer | Generative AI | Machine Learning

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Portfolio: pratiksinghds.github.io/Portfolio | LinkedIn: linkedin.com/in/pratiksingh-ds-aiml | GitHub: github.com/pratiksinghds

PROFESSIONAL SUMMARY

Entry-level AI/ML Engineer with a BSc in Data Science and an ongoing MBA (Data & AI specialization). Builds and deploys end-to-end Generative AI and machine learning applications, including Retrieval-Augmented Generation (RAG) pipelines, multi-agent LLM systems with LangGraph, recommendation engines and Power BI dashboards. Three projects are deployed as live web apps on Streamlit Cloud. Seeking an AI/ML Engineer, Data Scientist or Generative AI role to apply Python, LLM engineering and applied ML in production.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, Recommendation Systems, Clustering
- **Generative AI & LLMs:** LangChain, LangGraph, RAG, Multi-Agent Systems, Prompt Engineering, FAISS, Vector Databases, Hugging Face Embeddings, Groq API, Google Gemini API
- **Data & BI:** Power BI, DAX, PostgreSQL, Data Cleaning, Exploratory Data Analysis (EDA), Data Visualization
- **Tools & Deployment:** Streamlit, Streamlit Community Cloud, Git, GitHub, REST APIs, Linux, Jupyter Notebook

PROJECTS

Medical RAG Chatbot | *LangChain, Groq, FAISS, Hugging Face, Streamlit*

[Live Demo](#) | [GitHub](#)

- Built a domain-specific medical Q&A assistant that retrieves from a FAISS vector index of 7,000+ text chunks embedded with sentence-transformers (all-MiniLM-L6-v2).
- Reduced hallucinations with context-only prompting and page-level source citations shown for every answer.
- Deployed publicly on Streamlit Community Cloud with graceful handling of API rate limits.

Multi-Agent Travel Planning System | *LangGraph, Groq, PostgreSQL, Tavily, Streamlit*

[Live Demo](#) | [GitHub](#)

- Designed a four-agent LangGraph workflow (flight, hotel, itinerary and synthesis agents) that generates structured, budget-aware travel plans.
- Integrated real-time AviationStack flight data using LLM-based airport code extraction with fallback search, plus Tavily web search for hotels.
- Implemented PostgreSQL session checkpointing with an in-memory fallback for stateless cloud deployment.

Book Recommender System | *Scikit-learn, Pandas, Streamlit*

[Live Demo](#) | [GitHub](#)

- Built an item-based collaborative filtering engine using k-Nearest Neighbors on a user-book rating matrix, returning the five most similar titles with cover images.

Bank Data Analysis Dashboard | *Power BI, DAX, Python*

[GitHub](#)

- Cleaned and transformed real-world banking data and built DAX measures for Total Loan, Total Deposit, Business Lending and Credit Card Balance in an interactive dashboard.

Additional projects: Customer Churn Prediction (ANN, TensorFlow/Keras), Clustering Algorithms Study (K-Means, Hierarchical, DBSCAN), House Price EDA, Titanic Survival Prediction. [All projects](#)

EDUCATION

Master of Business Administration (MBA), Aviation Management

Sep 2025 – Jul 2027 (Expected)

IIMT University, Meerut | Specializations: Business Management, Data & AI

Bachelor of Science (BSc), Data Science

Sep 2022 – Aug 2025

IIMT University, Meerut

CERTIFICATIONS

- Machine Learning – E&ICT Academy, IIT Kanpur (MeitY initiative)
- Data Science – E&ICT Academy, IIT Kanpur (MeitY initiative)
- Linux – E&ICT Academy, IIT Kanpur (MeitY initiative)
- AI Fluency: Framework & Foundations – Anthropic (Claude Academy)
- Foundations of Prompt Engineering